

AI & Machine Learning on Google Cloud

Plain-language summary of the most important topics

Source: Introduction to AI and Machine Learning on Google Cloud (modules M0–M5 + AI Hypercomputer note)

1. What this course is about

AI can feel overwhelming and changes fast. This course turns curiosity into practical skills: from basic ideas, through generative AI and agents, to how you actually build and run models on Google Cloud—without needing to be a coding genius to get started.

Big idea: AI is like a powerful kitchen tool. Learn the ingredients and steps, then cook something useful—don't throw everything in a blender and hope.

2. AI foundations (Module 1)

Predictive vs generative AI

- **Predictive AI** answers questions like “What will happen?” (forecast sales, pick routes, classify data).
- **Generative AI** creates new content (text, images, video, code) and can also take actions through agents.
- Real businesses often need both. Example: Coffee on Wheels—routes and sales forecasts (predictive) plus marketing content automation (generative).

Google Cloud AI toolbox (three layers)

- **AI infrastructure** — compute, storage, networking, security (the foundation).
- **AI development** — platforms and tools to build models and agents.
- **AI applications & solutions** — ready products for end users.

Data → AI path

- **Ingest & process:** Pub/Sub, Dataflow, Spark, Data Fusion (batch and real-time).
- **Store & analyze:** Cloud Storage; SQL (BigQuery, Cloud SQL, AlloyDB, Spanner); NoSQL (Bigtable, Firestore); Looker for visualization.
- **Activate with AI:** Agent Platform and related build tools (Studio, AutoML, Notebooks).

How machines learn (basics)

- **AI** is the broad idea of machines doing smart tasks; **ML** is learning patterns from data.
- **Supervised learning:** labeled data, task-driven (you know the goal/answer).
- **Unsupervised learning:** unlabeled data, finds patterns (clustering, structure).
- Deep learning and generative AI sit on top of these ideas.

BigQuery ML (build models with SQL)

- Load/transform data into BigQuery.
- Select and preprocess features.
- Create the model in BigQuery (SQL).
- Evaluate how good it is.
- Use it to make predictions.

- Bonus: Gemini Code Assist can help explain, write, and debug SQL.

3. Generative AI (Module 2)

Gen AI creates multimodal content (text, code, images, speech, video, even 3D) from a **prompt** (instruction). It can summarize meetings, write reports, build chatbots—and with agents, book travel, automate workflows, and more.

Google's three-layer gen AI stack

- **Foundation models:** Gemini family (general), plus specialty models like Veo (video), Nano Banana (images), Embeddings.
- **AI development:** Model Garden, Agent Studio / Agent Builder, Agent Platform tools.
- **AI applications:** Gemini Enterprise app, Gemini Notebook (no-code / business-user entry points).

Prompt → production lifecycle

- A good prompt usually needs: **task**, **context**, and **examples**.
- Design → evaluate & refine → build & test → integrate & deploy → monitor & optimize.
- Useful knobs: temperature, top-K, top-P; compare prompts side by side.
- Improve accuracy with **grounding** and **RAG** (give the model your real company knowledge).
- Tuning options range from better prompts → parameter-efficient tuning → full fine-tuning.

AI agents (gen AI that takes action)

Agents go beyond chatbots: they pursue goals, call tools, and automate work.

- **Model (brain)** — reasoning and language.
- **Tools (hands/feet)** — connectors to systems and the outside world.
- **Orchestration (nervous system)** — process, workflows, coordination.

Pick tools by skill level: out-of-box Gemini Enterprise → templates / Agent Studio → pro-code Agent Development Kit (ADK).

4. AI development options (Module 3)

Google Cloud meets you where you are—for both generative and predictive AI:

- **No-code**: Gemini Enterprise app / for CX — out-of-box solutions.
- **No/low-code**: Agent Platform AutoML — train on your data with little/no ML coding.
- **Low-code**: Pre-trained APIs — ready models (vision, language, etc.) with no training data needed.
- **Code-based**: BigQuery ML, custom training (Python/JAX), ADK — full control DIY.

Agent Platform is the unified workspace to explore these options and build end-to-end ML projects.

Rule of thumb: start with the simplest option that solves the problem; go custom only when you need unique control or performance.

5. AI development workflow (Module 4)

Think of ML like running a restaurant kitchen:

- **1. Data preparation** — gather ingredients: upload data, engineer features. Data can be batch or streaming; structured or unstructured. Quality and quantity decide how well the model learns.
- **2. Model development** — cook and taste: train, then evaluate (does it work as expected?).
- **3. Model serving** — serve customers: deploy the model, then monitor and improve based on real use.

Two ways to build end-to-end

- **UI path** — e.g. Agent Platform AutoML (click-through workflow).
- **Code path** — SDKs and Agent Platform Pipelines for automation (CI / continuous training / delivery).

MLOps is the “back of house” that takes models from experiments into reliable production with monitoring and automation.

6. AI Hypercomputer (extra note)

AI Hypercomputer is Google’s integrated stack for heavy AI training and serving—hardware + software + flexible buying options.

- **Layer 1**: GPUs/TPUs, fast networking, optimized storage.
- **Layer 2**: open software (e.g. PyTorch, GKE, Kueue).
- **Layer 3**: consumption models (on-demand, spot, CUDs, reservations, DWS).

How you deploy it

- **Direct (GCE)** — max control, more ops work.
- **GKE** — balance of control and automation (Kubernetes).
- **Toolkits / open frameworks** — guided best-practice setups.
- **Fully managed (Vertex AI)** — easiest; Google runs more of the infra.

7. Cheat sheet — remember these

- AI = smart automation; ML = learn from data; Gen AI = create + (with agents) act.
- Google stack: infrastructure → development platforms → applications.
- BigQuery ML = ML with SQL for people who live in data warehouses.
- Prompts need task + context + examples; improve with evaluation, grounding/RAG, tuning.

- Agents = model + tools + orchestration.
- Build options: no-code → AutoML → pre-trained APIs → custom code.
- Workflow: prepare data → train/evaluate → deploy/monitor (then automate with MLOps).
- Best learning step after the course: apply one idea to your real work.

Sources summarized: M0_Course_introduction, M1_AI_foundations, M2_Generative_AI, M3_AI_development_options, M4_AI_development_workflow, M5_Summary, T-AIHYPE-I-m2-l1-en-file-8.en.pdf